

TechTutor SLM – Technology & Workflow Documentation

Core AI Technologies

TinyLlama-1.1B

TinyLlama is the base Small Language Model (SLM) used in this project. The model already understands natural language, and this project fine-tunes it into a coding tutor using custom datasets.

PyTorch

PyTorch is the deep learning framework responsible for running neural networks, tensor operations, GPU acceleration, and training workflows.

Hugging Face Transformers

The Transformers library is used to load pre-trained models, tokenizers, inference pipelines, and fine-tuning workflows.

Hugging Face Datasets

Used for loading, preprocessing, formatting, and managing custom instruction datasets during training.

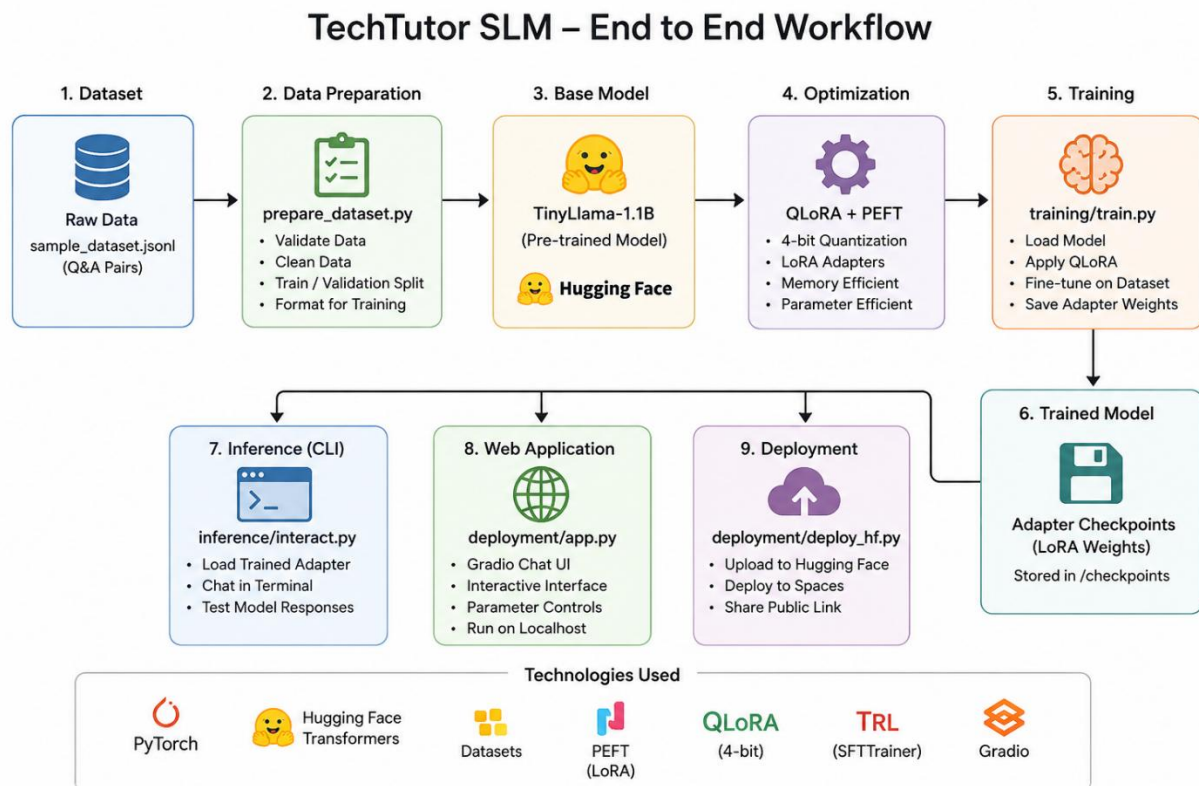
PEFT & QLoRA

PEFT (Parameter-Efficient Fine-Tuning) trains only lightweight adapter layers instead of the full model. QLoRA reduces GPU memory usage using 4-bit quantization for efficient training.

TRL (Transformer Reinforcement Learning)

Provides the SFTTrainer which simplifies supervised fine-tuning workflows for instruction-based datasets.

⚡ TechTutor SLM – Architecture Diagram



🐍 Python Scripts & Workflow

data/prepare_dataset.py

This script processes the raw dataset and prepares it for training.

Functions:

- Reads JSONL datasets
 - Validates dataset structure
 - Splits dataset into Train & Validation sets
 - Converts conversations into model-friendly format
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training/train.py

This is the main model training script.

Functions:

- Loads TinyLlama model
 - Applies QLoRA optimization
 - Configures LoRA adapters
 - Starts supervised fine-tuning
 - Saves adapter checkpoints after training
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inference/interact.py

This script is used for local CLI-based testing.

Functions:

- Loads trained adapter model
 - Accepts terminal-based prompts
 - Generates AI responses in real time
 - Tests model performance locally
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deployment/app.py

This script creates the web application using Gradio.

Functions:

- Builds chatbot UI
- Connects frontend with trained model

- Provides chat interaction in browser
- Supports parameter tuning like temperature and max tokens

Run locally:

```
python deployment/app.py
```

Open:

```
http://localhost:7860
```

deployment/deploy_hf.py

This script deploys the project to Hugging Face Spaces.

Functions:

- Uploads adapter checkpoints
 - Pushes project to Hugging Face Hub
 - Creates public AI demo link
 - Hosts Gradio application online
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Environment & Setup

requirements.txt

Contains all Python dependencies required for the project.

Libraries Included:

- torch
- transformers
- peft
- datasets
- trl
- gradio
- accelerate
- bitsandbytes

Install all dependencies:

```
pip install -r requirements.txt
```

Virtual Environment (venv)

A separate isolated Python environment used to avoid dependency conflicts between projects.

Create environment:

```
python -m venv venv
```

Activate environment:

```
venv\Scripts\activate
```

Complete Workflow

1. Prepare dataset using prepare_dataset.py
 2. Load TinyLlama model
 3. Apply QLoRA optimization
 4. Fine-tune using train.py
 5. Test using interact.py
 6. Launch chatbot UI using app.py
 7. Deploy publicly using deploy_hf.py
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Project Goal

The main objective of this project is to understand Small Language Model architecture, fine-tuning workflows, adapter training, inference pipelines, and AI deployment using modern open-source AI technologies.